Mini Paper — 1.4 Data Types, Data Structures & Boolean Algebra

Time: ~35 minutes · Total: 30 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark. Show all working — method marks are available.

Questions

Q1. Convert the denary number **150** into: (a) 8-bit binary **[1]** (AO2)

(b) hexadecimal **[2]** (AO2)

Show your working.

Q2. Convert the hexadecimal number **2F** into denary. Show your working. **[2]** (AO2)

Q3. Using **8-bit two's complement**, calculate **45 − 58**. You must: (a) show 45 and 58 in 8-bit binary **[1]** (AO2)

(b) form the two's complement of 58 **[1]** (AO2)

(c) perform the addition and state the final denary result **[2]** (AO2)

[4 total]

Q4. A floating point value is shown as `0.00110 × 2³` (mantissa with the point after the sign bit, exponent in denary). (a) Normalise the mantissa and state the new exponent. **[2]** (AO2)

(b) Explain *why* numbers are normalised. **[1]** (AO1)

[3 total]

Q5. The 8-bit signed value `1111 0100` represents **−12** in two's complement. (a) Apply an **arithmetic shift right by 1** and write the resulting bit pattern. **[1]** (AO2)

(b) State the denary value of the result and explain why an *arithmetic* (not logical) shift is required here. **[2]** (AO2)

[3 total]

Q6. Simplify the Boolean expression **$\neg(A + \bar{B}) + A \cdot B$** using De Morgan's law and Boolean identities. Show each step. **[4]** (AO2)

Q7. Complete the truth table for the expression **F = A·B̄ + Ā·B** and state which single logic gate it is equivalent to. **[3]** (AO2)

A	B	F
0	0	
0	1	
1	0	
1	1	

Q8. A queue is implemented as a **circular queue** in an array of size 5 (indices 0–4), initially empty with `front = 0` and `rear = 0`. The operations below are applied in order: enqueue P, enqueue Q, enqueue R, dequeue, enqueue S, enqueue T. (a) Show the array contents and the values of `front` and `rear` after all operations. **[3]** (AO3)

(b) State **one** advantage of a circular queue over a linear queue. **[1]** (AO1)

[4 total]

Q9. The values **50, 30, 70, 20, 40, 60** are inserted, in that order, into an initially empty **binary search tree**. (a) Draw or describe the resulting tree. **[2]** (AO3)

(b) State the order in which the values are visited by an **in-order** traversal, and what property of the BST this demonstrates. **[2]** (AO3)

[4 total]